

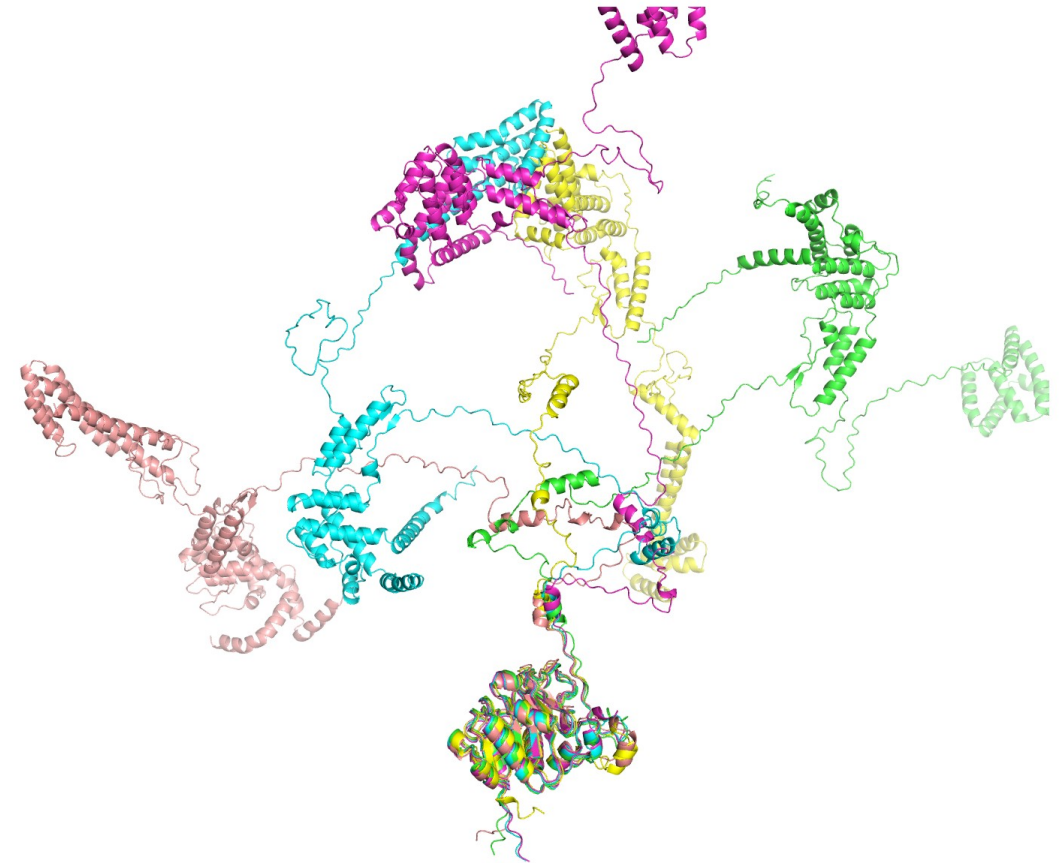
IFT52:IFT46, Part 2: did they really bind?

Let us go through your complex predictions together

Cilia Connect 2026 · AI Structure Prediction Workshop

The five models agreed on just one region

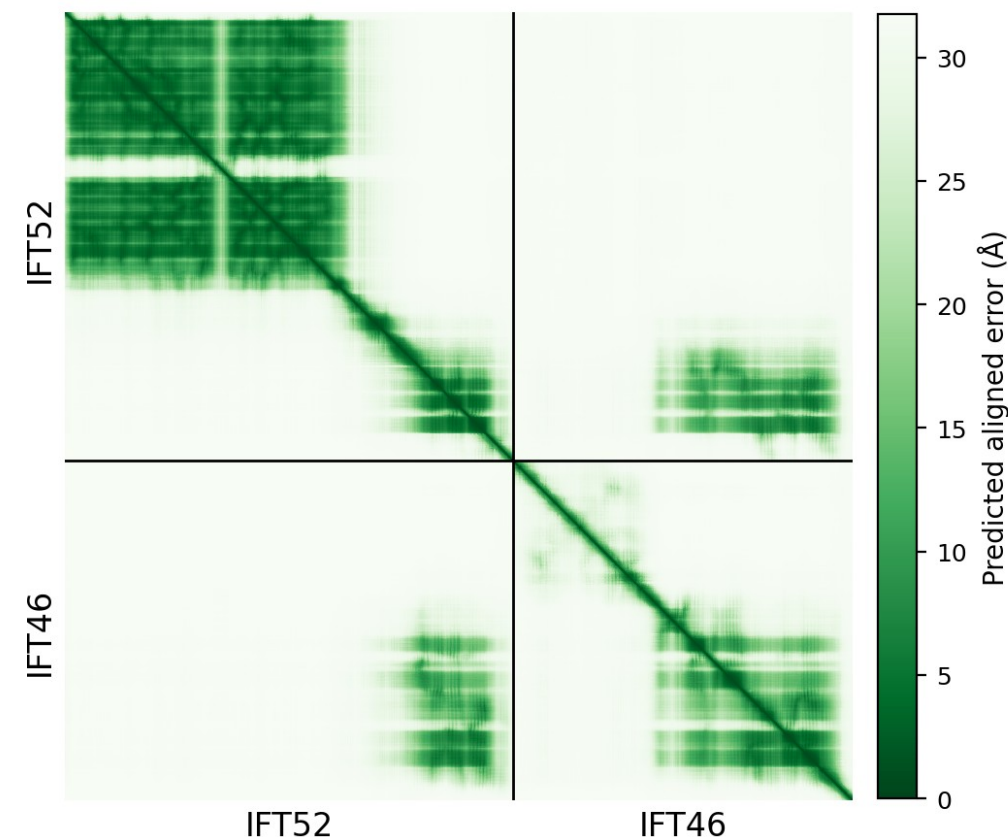
- You superposed all five independent predictions. Across most of each protein the colours fan apart: those parts land somewhere different every time.
- One compact region sits in the same place in all five models: the two C-terminal halves, packed against each other.
- Agreement across independent models is AlphaFold telling you it is confident. That one reproducible region is your candidate interface.



Five models, one colour each; only the C-terminal contact lands on top of itself.

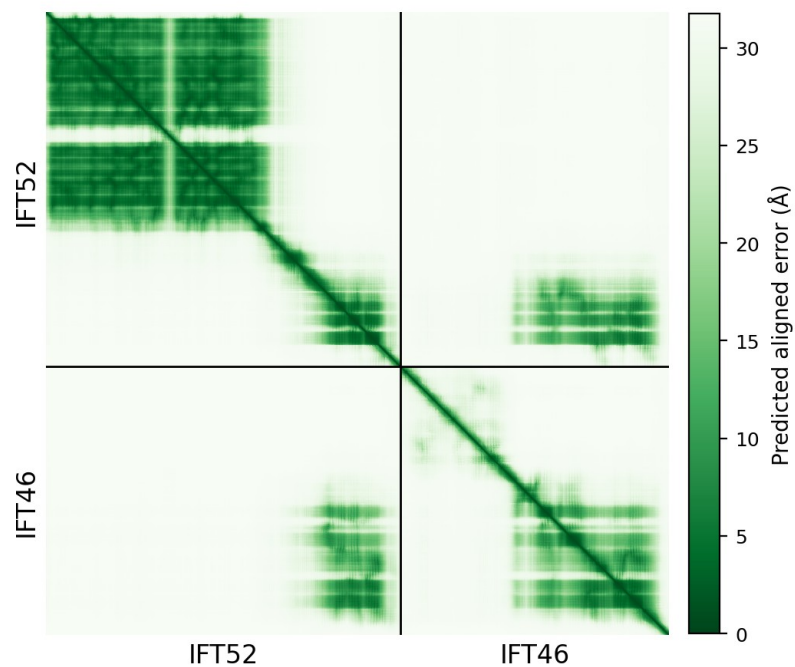
A real interface, hidden behind a low ipTM

- The PAE has one dark off-diagonal block, exactly where the five models agreed: the IFT52 and IFT46 C-termini.
- Everything else is pale. The long non-interacting halves float freely relative to each other.
- ipTM = 0.35. But ipTM averages over all residue pairs between the chains, so the floppy majority buries the real interface.



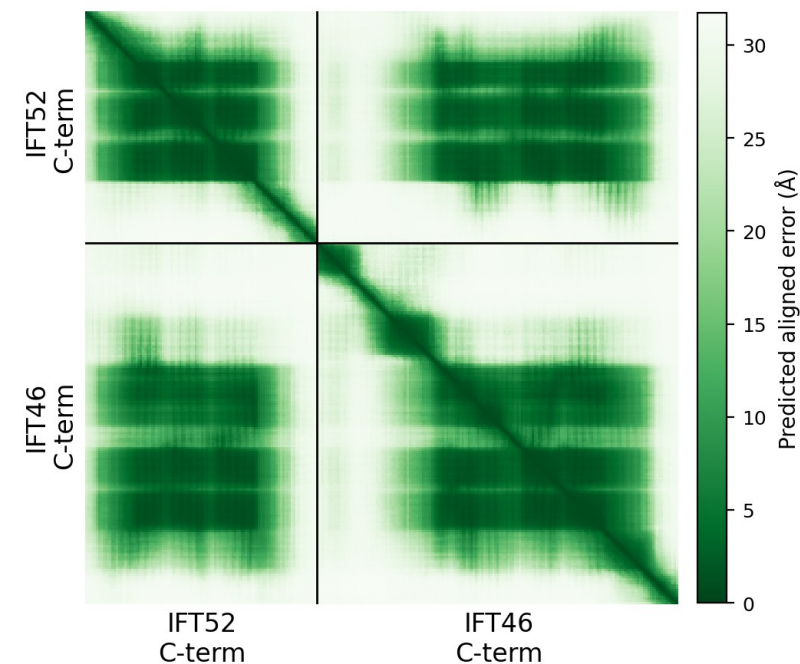
A low ipTM did not mean no interaction. It meant mostly disordered, with a genuine interface inside: min interface PAE 3.96 Å, interface pLDDT ~71.

Test it: cut away the floppy halves



Full length ipTM 0.35

Re-predict only the
interface

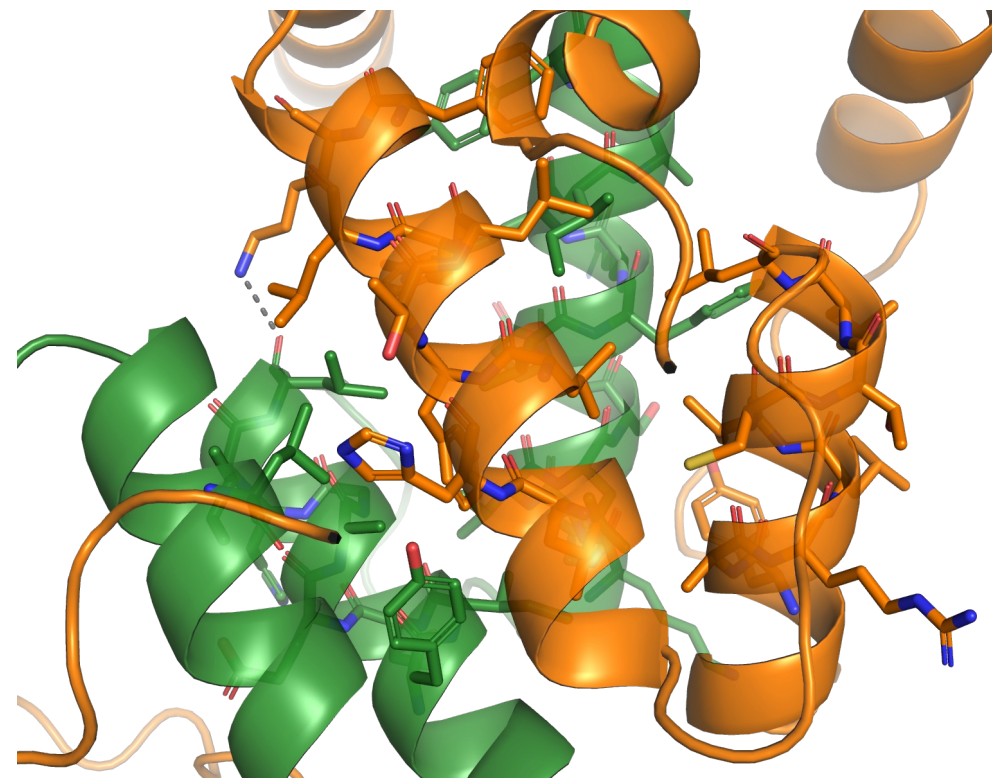


C-terminal only ipTM 0.65

The interface was real all along. ipTM 0.35 → 0.65 · min interface PAE 3.96 → 0.95 Å · interface pLDDT 71.5 → 89.6. Removing the diluting halves let the score reveal it.

Validation: this is the known interface

- The contact you pinned down: IFT52 residues ~357–436 against IFT46 ~181–322, the C-terminal regions of both proteins.
- IFT52 and IFT46 are known from biochemistry and crystallography to bind through exactly these C-terminal regions.
- AlphaFold reproduced, from sequence alone, a complex we already had experimental evidence for.



Why this matters: a method that recovers a known interface earns your trust on the next one, a complex nobody has solved yet.

Part 2 take-home messages

- pLDDT is local and cannot tell you whether two chains bind. The PAE and ipTM describe relative position and interfaces.
- ipTM is a global average, so a real but localized interface can hide behind a low ipTM.
- Read the PAE: look for a dark inter-chain block. Then test it, truncate to the interface and re-predict.
- Reproducing a known interaction is how you validate the method before trusting it on an unknown one.
- You saw a clean complex (IFT25:27, ipTM 0.91) and a hard one (IFT52:46, ipTM 0.35 yet real). The number alone would have misled you on the second.

The one rule: never judge a predicted complex by ipTM alone.